Speech Recognition

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***Abstract*—In this lab, we created a speech recognition system that could recognize and classify a spoken word from a given dictionary of words. Additionally, if the word was not in the dictionary, the system would return that the word was not in the dictionary. We extended this project by including a second speaker and gave the system the capability to discern between the two different speakers and then determine which word was spoken. The system was correctly able to identify the spoken word 84% of the time when there was only one speaker in ideal conditions. The accuracy fell when extending to two speakers; however, the system was able to predict the correct speaker 97% of the time.**

I. Introduction

Modern speech recognition techniques typically utilize internet-connected analysis and deep learning models that require significant training. These models, at their core, rely on spectral attributes of speech. The main goal of this project was to achieve real-time speech recognition using basic audio signal processing. The main objectives of the project were the following: characterize the audio recording devices that we utilized, write a function that creates a spectrogram in real-time from a real-time audio signal, develop the ability to recognize speech and classify words from a created dictionary, and to recognize who was speaking from two different speakers and identify what word they said.

II. Methods

Each objective had its own various methods that when put together, achieved the overall goal of the project.

A. Characterization of Audio Devices

Audio devices typically employ some kind of signal processing on the audio signals that they handle. It is important to characterize how these audio devices react to different signals to ensure that they act as expected when we built our entire speech recognition system. The first metric of characterization that we explored was the noise inherent in the acquisition card that we utilized. This was accomplished using a loop-back cable that connected the input and output ports together and measured the synchronized input/output signal.

The signals in figure 1 were sampled at a rate of 44.1 kHz for 5 seconds. The input noise level was determined to be -24 dB and the output noise level was determined to be -65 dB, meaning that the noise in the sound card would not introduce significant noise when we would eventually record audio for speech recognition.

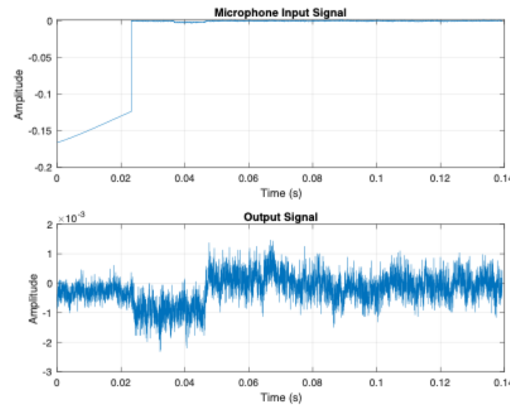

Fig. 1. Input and output signals of acquisition card when input and output ports are connected together

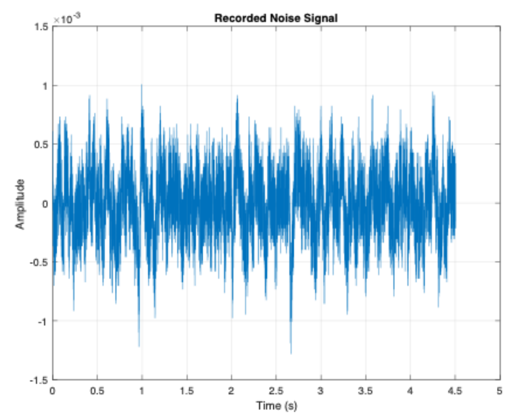

Fig. 2. Output signal when there is no input connected

We additionally assessed the noise when there was no input or output to better characterize the acquisition card.

The signal in figure 2 was again sampled at a rate of 44.1 kHz for 5 seconds. The noise level was calculated to be -72 dB.

The other metric of characterization that we explored was the frequency response of each audio device to see how the different audio devices reacted to different signals. The frequency response of a system describes the change of magnitude of certain frequencies by the system.

To measure the frequency response of the sound card, we again connected the input and output ports of the sound card.

We then swept through sine waves of various frequencies through the input and measured the output signal. To obtain the frequency response of a signal, one must divide the Fourier Transform of the output signal by the Fourier Transform of the input signal. The following figure depicts the frequency response of the sound card:

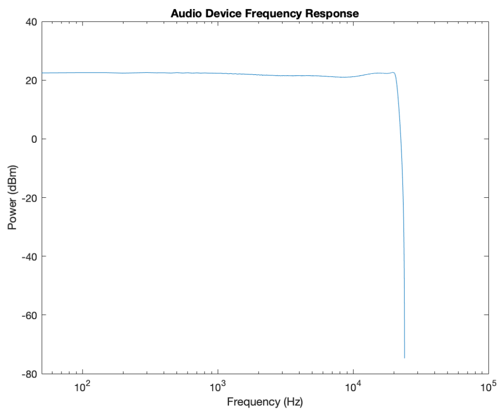

Fig. 3. Frequency response of sound card

As seen in figure 3, the sound card boosts all sounds that are passed through it just until the end of the sweep where the sound card then massively attenuates the signal.

The next frequency response we characterized was a 2019 MacBook Air's built-in microphone. We conducted the same sweep of sine waves as in the sound card example and played them through the MacBook's built-in speakers and recorded through the built-in microphone.

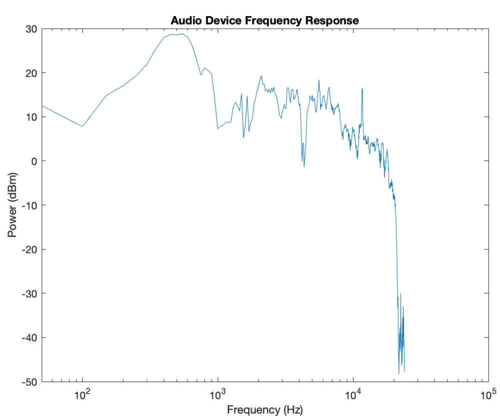

Fig. 4. Frequency response of built-in MacBook microphone

As seen in figure 4, the MacBook Air's built-in microphone adds a gain factor of between 10 and 30 dB all the way up to about 10,000 Hz. From there, the microphone attenuates all signals until the simulation ends.

The final frequency response that we characterized was the provided Audio Technica microphone.

As seen in figure 5, the Audio Technica microphone attenuates all frequencies by some degree. After 10,000 Hz this attenuation increases greatly.

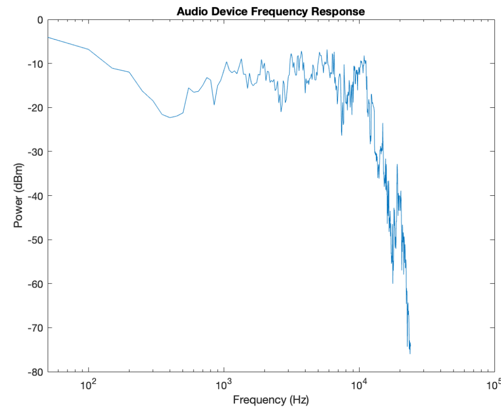

Fig. 5. Frequency response of Audio Technica microphone

Now that all of our audio devices had been characterized, we were able to begin building our spectrogram function.

B. Spectrogram Function

A spectrogram has three components: time, frequency, and amplitude. Time is represented on the horizontal axis, frequency is represented on the vertical axis, and the amplitude is represented by color. For each point in time, the spectrogram shows how much of each frequency was present. This is important for speech recognition because when a person says a word, each part of the word is made up of different frequencies, so knowing when each frequency occurred is essential for identifying the word. If only the Fourier transform of the audio is considered, then the timing of each frequency would be lost.

The first step in producing a spectrogram in real-time was recording audio and adding it to the buffer. The audio was recorded by sampling it at 48 kHz for 5 seconds. The buffer was used to store the audio until it could be processed. We found that the average syllable takes around 0.2 seconds to say, so we needed to read at least 9,600 samples at a time in order to increase the likelihood of processing the whole syllable at once. We also decided to overlap 3/4 of the samples to prevent losing information between readings. Using these two values, we determined that we had to read 16,800 samples at a time in order to keep syllables together and overlap consecutive readings.

Since we would eventually take the fft of each reading, we had to consider what the segment looked like in the frequency domain. Each segment was finite in time, which means it would be infinite in the frequency domain. We are not able to analyze an infinite number of frequencies, so we only looked at a finite range of frequencies, leading to a loss of information. In order to reduce this loss, we considered different windowing techniques. Directly processing each reading from the buffer is equivalent to multiplying the buffer by a rectangular pulse. The Fourier transform of a rectangular pulse has many side lobes, which would lead to the loss of a lot of information. To improve this, we multiplied each reading from the buffer with a triangular pulse. This pulse is still infinite in the frequency domain, but it has smaller side lobes, so less information would be loss than if we used a rectangular pulse. As shown in figure 6, the triangular pulse created a spectrogram with clearer details and less blurring.

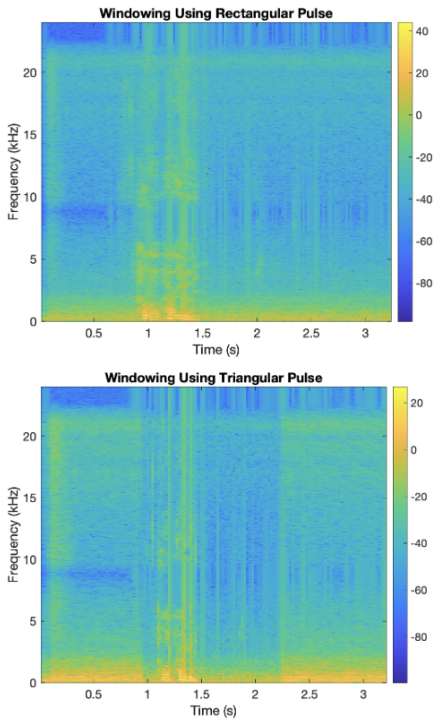

Fig. 6. Comparison between rectangular pulse and triangular pulse for windowing

After multiplying the buffer reading by the triangular pulse in the time domain, we took the fft in order to analyze the signal in the frequency domain. Each array we got after taking the fft corresponded to one column of the spectrogram. From here, we used the imagesc() command to plot time vs. frequency with the color of the graph representing the amplitude.

C. Baseline Speech Recognition

Our speech recognition model used the information in the spectrogram to determine which word was spoken. The first step for completing this was to create a dictionary. Our dictionary consisted of 5, 2 syllable words: tiger, monkey, panda, giraffe, and lion. For each word in the dictionary, we recorded one of us saying the word and extracted just the word from the audio. Isolating the word allowed us to compare spectrograms since only information about the word would be shown. To do this, we calculated the signal to noise ratio for each point in time until we found a time where the SNR is greater than 25 dB. Once the starting index of the word was known, we kept the next 1 second of audio since this amount of time ensured the whole word would be included. Next, we created the spectrogram for each word, and each spectrogram was stored in the dictionary.

When a new recording was completed to test the accuracy of our speech recognition system, we followed similar steps as when making the dictionary. First, the word is isolated in the recorded audio, and then the spectrogram is created. When isolating the word, if there is no time in the recording where the signal to noise ratio is greater than 25 dB, then our speech recognition system returns that no word was said. If the signal to noise ratio was greater than 25 dB at some point, the new spectrogram is correlated with each of the spectrograms in the dictionary. The word that returned the highest correlation coefficient is selected as the spoken word. The chosen word must have a correlation coefficient greater than 0.7. Otherwise, the system returns that the recorded word is not in the dictionary.

D. Distinguishing Between Two Different Speakers

In order to identify the human speaker, we had to use a larger dictionary that contained each word spoken by each person. We tested speaker identification with two human speakers, so the updated dictionary consisted of 10 words. Again, when a person spoke into the microphone, the word was isolated using the threshold for the signal to noise ratio, and then a spectrogram was created. The correlation coefficient between the recording and each of the 10 words in the dictionary was used to identify which word was said by which person.

III. Results

Our baseline speech recognition system was able to correctly identify the spoken word 84% of the time, given that a word in the dictionary was said under ideal conditions. The confusion matrix for this test is shown in figure 7. When we tested the system but added noise and had the speaker stand farther away from the microphone, the accuracy decreased to 52%.

		Actual Word				
		Tiger	Monkey	Panda	Giraffe	Lion
Predicted Word	Tiger	4	0	1	0	0
	Monkey	1	4	1	0	0
	Panda	0	1	3	0	0
	Giraffe	0	0	0	5	0
	Lion	0	0	0	0	5

Fig. 7. Confusion matrix when 1 speaker said spoke during ideal conditions

As shown in figure 8, the system was typically able to correctly identify if a word was spoken since the signal to noise ratio was very different depending on if someone was speaking or not. However, it was not a perfect measurement since if the person wasn't talking at a high enough volume, then the word was not detected.

Our system was not able to reliably determine if the spoken word was in the dictionary or not, as shown in figure 9. The correlation coefficients between any two words was high, and the range of the coefficients varied depending on the recording conditions. Because of this, we were unable to find a good threshold for the correlation coefficients to decide if the spoken word was in the dictionary or not.

		Actual	
		Word	Not Word
Prediction	Word	4	0
	Not Word	2	6

Fig. 8. Confusion matrix when detecting if a word was spoken

		Actual	
		In Dictionary	Other
Prediction	In Dictionary	5	3
	Other	1	3

Fig. 9. Confusion matrix when deciding if the spoken word is in the dictionary or not

Our speaker identification system was typically able to correctly predict detect who was speaking, but the accuracy for the second speaker was not as high as it was for the first speaker. The confusion matrix is shown in figure 10, and in this figure, the letter is the first letter of the spoken word, and the number corresponds to which speaker was talking (ex: T_1 means speaker 1 said "tiger"). Our system was able to predict the correct speaker 97% of the time, which was likely because the two speakers had voices that sounded very differently. The lower accuracy for the second speaker suggests that our system may have been over-fitted for the first speaker, since we tested the baseline system using speaker 1's voice.

		Actual Word									
		T_1	M_1	P_1	G_1	L_1	T_2	M_2	P_2	G_2	L_2
Predicted Word	T_1	2	0	0	0	0	0	0	0	0	0
	M_1	1	3	1	0	0	0	0	0	0	0
	P_1	0	0	1	0	0	0	0	0	0	0
	G_1	0	0	1	2	0	0	0	0	1	0
	L_1	0	0	0	1	3	0	0	0	0	0
	T_2	0	0	0	0	0	0	0	1	0	0
	M_2	0	0	0	0	0	1	2	1	0	0
	P_2	0	0	0	0	0	0	0	1	0	1
	G_2	0	0	0	0	0	2	1	0	0	0
	L_2	0	0	0	0	0	0	0	0	2	2

Fig. 10. Confusion matrix when deciding which speaker said what word

IV. Discussion and Conclusions

. Our system was not as accurate as we had hoped when classifying words from speaker 2. As previously mentioned, we suspect this to be due to overfitting. It would be a great use of time to go back and determine precisely why this drop off in performance occurred. For example, we could inspect all the dictionary spectrograms and compare them to the ones that that we are passing through.

Specifically, we should review the dictionary spectrogram for "panda" since this was the word that our system frequently misclassified. Figure 11 shows the spectrogram of an audio recording of "panda" that was misclassified as "tiger". Figure 12 shows the dictionary spectrogram for "panda". These spectrograms are clearly different, so it would be beneficial to identify another metric that could be used to classify words.

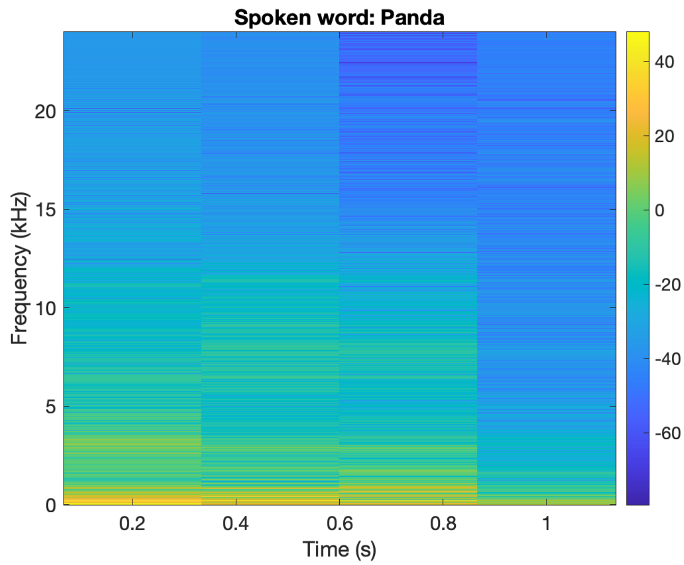

Fig. 11. Spectrogram of the audio when "panda" was misclassified.

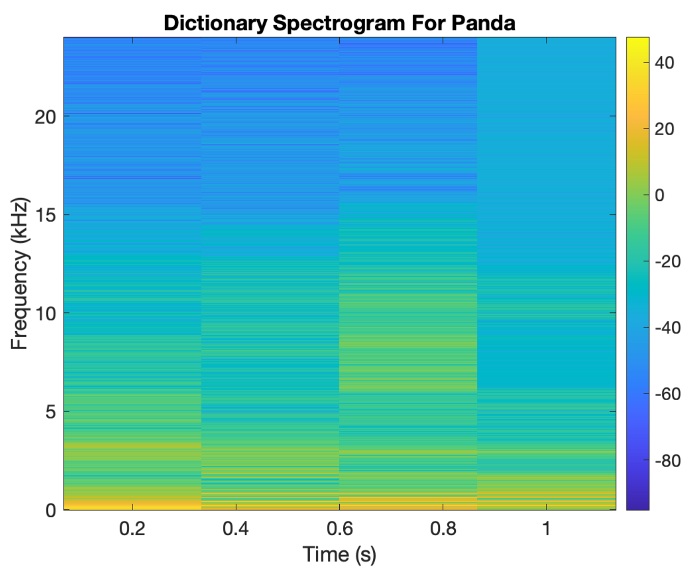

Fig. 12. Dictionary spectrogram of "panda"

As stated in the previous section, our system was very accurate when discerning between the two speakers. We suspect this to be because that the speaker's voices vary greatly. Speaker 2's voice is noticeably deeper than Speaker 1's. It would be worthwhile to test the system with speakers that have similar voices to truly determine how robust our system is with discerning speakers.

Another area of further research, would be in tuning the system to work on a wider range of speakers. The system has a significant amount of parameters including but not limited to: sample rate, correlation coefficient threshold, spectrogram window type, power threshold to determine if a word was spoken, and many more. The system could be improved if we ran an iterative analysis on how each parameter affected recognition and then determined which sets of parameters gave us the accuracy for both speaker and word detection. This would hopefully make the system speaker agnostic meaning that it can classify words from the dictionary no matter who the speaker is.

Through basic signal and systems concepts, we were able to build a speech recognition system that was able classify words from a dictionary at a high accuracy. We extended the project to include a second speaker and adjusted the system so that it could discern between speakers and classify words.